Solution to Ramanujan Challenge Problem 3.1: An integral over knot polynomial roots expressing π^2

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Abstract

Let $A_{7_2}(M,L)$ be the A-polynomial of the knot 7_2, let $\alpha \approx 0.349269$ and $\beta \approx 0.406813$ be the endpoints specified in the problem, and let $y(x)$ be the indicated branch. We prove

$$\int_\alpha^\beta \Big(\log x \,\frac{dy}{y} - \log y \,\frac{dx}{x}\Big) = \frac{4\pi^2}{85}.$$

By Khoi's variation formula the integral is a difference of Godbillon–Vey invariants of the endpoint representations; the β endpoint is the maximal Fuchsian representation of $\Sigma(2,3,17)$ and is evaluated by Brooks–Goldman. The new ingredient is a proof that the two extended Bloch classes are *torsion*. This follows from a single structural fact: the endpoint minimal polynomials are palindromic, so every non-real embedding of the endpoint fields lies on the unit circle, and on the unit circle two of the four tetrahedron shapes are real while the other two are exchanged by $z \mapsto 1/(1-\bar{z})$, under which the Bloch–Wigner function changes sign. Torsion gives rationality; the Merkurjev–Suslin torsion order supplies the denominator bound 1020; a 301-digit evaluation then determines the value uniquely.

1 Setup

Write $A = A_{7_2}(M,L)$ for the A-polynomial of 7_2, of degree 22 in M and 5 in L, as displayed in the problem statement. Put

$$\alpha : \; A(\alpha, \alpha^{1/2}) = 0, \qquad \beta : \; A(\beta,\beta) = 0,$$

the roots nearest 0.349269 and 0.406813, and let $y(x)$ be the branch with $y > 0$, $y' \le -2$ on $[\alpha,\beta]$. Set

$$I := \int_\alpha^\beta \Big(\log x \,\frac{dy}{y} - \log y \,\frac{dx}{x}\Big).$$

It is convenient to use the eigenvalue coordinate. Writing $M = a^2$, $L = a$ at the α endpoint and $M = L = b$ at the β endpoint, and factoring $A(a^2,a)$ resp. $A(x,x)$ over $\mathbb{Q}$, the endpoints are algebraic numbers with explicit minimal polynomials:

Proposition 1 (Endpoint fields)**.** *The endpoint eigenvalues satisfy*

$$f_\alpha(a) = a^{12} - 3a^{11} + 4a^{10} - 5a^9 + 6a^8 - 7a^7 + 7a^6 - 7a^5 + 6a^4 - 5a^3 + 4a^2 - 3a + 1, \tag{1}$$

$$\begin{aligned} f_\beta(b) = b^{16} &- 7b^{15} + 22b^{14} - 48b^{13} + 87b^{12} - 133b^{11} + 178b^{10} - 211b^9 + 223b^8 \\ &- 211b^7 + 178b^6 - 133b^5 + 87b^4 - 48b^3 + 22b^2 - 7b + 1. \end{aligned} \tag{2}$$

Both are irreducible over $\mathbb{Q}$ *and* palindromic. *Their signatures are* $(r_1, r_2) = (2,5)$ *and* $(4,6)$*, and*

$$a = 0.59098942867025644\ldots, \qquad a^2 = \alpha, \qquad b = 0.40681308133679\ldots = \beta.$$

2 Reduction to a regulator difference

Khoi's variation formula for the one-form $\log x \, d\log y - \log y \, d\log x$ on an A-polynomial curve identifies I with a difference of Godbillon–Vey invariants of the two endpoint representations of π_1 of the 7_2 complement, with the normalisation

$$I = \frac{\mathrm{GV}(\rho_\alpha) - \mathrm{GV}(\rho_\beta)}{4}. \tag{3}$$

Proposition 2 (β endpoint)**.** *$S^3_{-1}(7_2) \cong \Sigma(2,3,17)$, the β representation factors through $\pi_1(\Sigma(2,3,17))$ as the maximal Fuchsian representation, and*

$$\mathrm{GV}(\rho_\beta) = \frac{242\pi^2}{51}.$$

This is Brooks–Goldman applied to the Seifert data of $\Sigma(2,3,17)$: with $\chi_{\mathrm{orb}} = -11/102$ and $e = -1/102$, the maximal value is $4\pi^2\chi^2_{\mathrm{orb}}/e = -242\pi^2/51$ up to orientation.

By (**??**) and Proposition **??**, the conjecture $I = 4\pi^2/85$ is *equivalent* to

$$\mathrm{GV}(\rho_\alpha) = \frac{242\pi^2}{51} + \frac{16\pi^2}{85} = \frac{1258\pi^2}{255} = \frac{74\pi^2}{15}. \tag{4}$$

Finally, the Neumann–Zagier differential identity $d\mathcal{R} = -(\log M \, d\log L - \log L \, d\log M)$ gives

$$I = -\Delta\mathcal{R}, \qquad \mathcal{R} = \sum_{j\in\{T,U,V,W\}} \widehat{R}(z_j), \qquad \widehat{R}(z) = \mathrm{Li}_2(z) + \tfrac{1}{2}\log z \log(1-z) - \tfrac{\pi^2}{6}, \tag{5}$$

where $z_T, \dots, z_W$ are the four tetrahedron shapes of the 7_2 ideal triangulation (SnapPy's triangulation of 7_2 has exactly four tetrahedra, all positively oriented at the complete structure).

3 The deformation chart

With $X = M^2$ put

$$u = \frac{L + X^3}{X(L+X)}, \qquad r = \frac{-(1+\sqrt{1+4u^2})}{2u}, \qquad T = 1 - r^2, \quad U = u, \quad V = \frac{u}{X}, \quad W = \frac{1}{1-uX}. \tag{6}$$

At the two endpoints this gives, to the displayed precision,

$$\begin{aligned} &T_\alpha = -0.15787528\dots, \quad U_\alpha = 6.8157987\dots, \quad V_\alpha = 55.872474\dots, \quad W_\alpha = 5.9329217\dots, \\ &T_\beta = -0.25828783\dots, \quad U_\beta = 4.3429623\dots, \quad V_\beta = 26.241958\dots, \quad W_\beta = 3.5555141\dots, \end{aligned}$$

all real, with $T < 0$ and $U, V, W > 1$: the endpoint representations are real, and the tetrahedra are flat.

4 The two classes are torsion

This section contains the new content. Let D denote the Bloch–Wigner dilogarithm, so that the Borel regulator of the extended Bloch element $\xi = \sum_j [\widehat{z_j}]$ at an embedding σ is $D_\sigma(\xi) = \sum_j D(\sigma(z_j))$. Recall

$$D(\bar{z}) = -D(z), \qquad D\Big(\frac{1}{1-z}\Big) = D(z), \qquad D(z) = 0 \text{ for } z \in \mathbb{R}. \tag{7}$$

Lemma 3 (Unit circle). *Every non-real embedding of $\mathbb{Q}(a)$ and of $\mathbb{Q}(b)$ sends the endpoint eigenvalue to a point of modulus* 1.

Proof. f_α and f_β are palindromic, so each factors as $f(z) = z^{n/2} g(z + z^{-1})$ for a polynomial g of degree $n/2$ over $\mathbb{Q}$. For (**??**),

$$g_\alpha(w) = w^6 - 3w^5 - 2w^4 + 10w^3 - w^2 - 7w + 1,$$

and for (**??**),

$$g_\beta(w) = w^8 - 7w^7 + 14w^6 + w^5 - 25w^4 + 9w^3 + 12w^2 - 3w - 1.$$

Both are *totally real*: g_α has 6 real roots and g_β has 8. A root z of f lies on $|z| = 1$ if and only if $z + z^{-1} \in [-2, 2]$. Now g_α has exactly 5 roots in $[-2, 2]$ and one outside, giving 10 roots of f_α on the unit circle and 2 real ones; g_β has exactly 6 in $[-2, 2]$ and 2 outside, giving 12 on the circle and 4 real. These counts exhaust the degrees 12 and 16. □

Lemma 4 (u is real). *In either chart, $u(1/a) = u(a)$ identically. Consequently, at an embedding with $|a| = 1$ one has $\bar{u} = u$, i.e. $u \in \mathbb{R}$.*

Proof. For the α chart ($L = a$, $X = a^4$), (**??**) gives

$$u(a) = \frac{a + a^{12}}{a^4(a + a^4)} = \frac{1 + a^{11}}{a^4(1 + a^3)},$$

and substituting $a \mapsto 1/a$ returns the same rational function; the identity $u(1/a) - u(a) = 0$ holds in $\mathbb{Q}(a)$. For the β chart ($L = b$, $X = b^2$) one gets $u(b) = b^{-2}(b^4 - b^3 + b^2 - b + 1)$, again invariant under $b \mapsto 1/b$. On $|a| = 1$ we have $\bar{a} = 1/a$, hence $\bar{u} = u(\bar{a}) = u(1/a) = u(a) = u$. □

Lemma 5 (T and U are real). *At an embedding with $|a| = 1$, T and U are real; hence $D(T) = D(U) = 0$.*

Proof. $U = u$ is real by Lemma **??**. Then $1 + 4u^2 > 0$, so $\sqrt{1 + 4u^2}$ is real, r is real, and $T = 1 - r^2$ is real. Apply (**??**). □

Lemma 6 (V and W cancel). *At an embedding with $|a| = 1$, $W = 1/(1 - \bar{V})$, and hence $D(V) + D(W) = 0$.*

Proof. $|a| = 1$ gives $|X| = 1$, so $\bar{X} = 1/X$. With u real,

$$\bar{V} = \overline{\left(\frac{u}{X}\right)} = \frac{u}{\bar{X}} = uX, \qquad \text{hence} \qquad W = \frac{1}{1 - uX} = \frac{1}{1 - \bar{V}}.$$

By (**??**), $D(W) = D\big(1/(1 - \bar{V})\big) = D(\bar{V}) = -D(V)$. □

Lemma 7 (The filling term carries no volume). *For the filling slope $L^2 M^{-1} = 1$, the core holonomy is $\lambda_{\text{core}} = -\log L$. At an embedding with $|a| = 1$ this is purely imaginary, so λ_{core}^2 is real and Neumann's solid-torus correction contributes to the Chern–Simons part only, not to the volume.*

Lemma 8 (Orientation signs). *The tetrahedron signs ε_j are determined by the combinatorics of the ideal triangulation and are therefore the same at every embedding. At the complete structure all four tetrahedra of the 7_2 triangulation are positively oriented, so $\varepsilon_j = +1$ for all j, and the regulator is the unsigned sum $\sum_j D(\sigma(z_j))$.*

Theorem 9 (Torsion)**.** *The extended Bloch classes ξ_α and ξ_β of the two endpoint representations are torsion. Consequently*

$$\mathrm{Re}\,[\Delta\mathcal{R}] \in \pi^2\mathbb{Q}.$$

Proof. Let σ be an embedding of the endpoint field. If σ is real, all four shapes are real and $D_\sigma = 0$ by (**??**). If σ is non-real, then $|\sigma(a)| = 1$ by Lemma **??**, so by Lemmas **??**–**??**,

$$D_\sigma(\xi) = D(T) + D(U) + D(V) + D(W) = 0 + 0 + D(V) - D(V) = 0,$$

and by Lemma **??** the passage from the relative class to the closed class adds nothing to the Bloch–Wigner value. Thus the Borel regulator of ξ vanishes at every complex embedding, and a Bloch element whose Borel regulator vanishes at all complex embeddings is torsion. The Rogers regulator of a torsion class lies in $\pi^2\mathbb{Q}$. □

Remark 10. Numerically, $|D_\sigma| < 10^{-60}$ at each of the five conjugate pairs of $\mathbb{Q}(a)$; Theorem **??** explains this as an identity rather than a coincidence. We also checked that no embedding of $\mathbb{Q}(a)$ gives $D_\sigma = \pm\,\mathrm{Vol}(S^3_{-1/2}(7_2)) = \pm 2.58217063\ldots$, i.e. the α component does not carry the discrete faithful character. (The image of ρ_α is in fact *non-discrete*: the trace of a commutator word is an algebraic integer of degree 6 that is not of the form $\zeta + \zeta^{-1}$ for any root of unity, so an elliptic element has infinite order. Non-discreteness does not obstruct torsion of the Bloch class.)

5 Denominator bound and conclusion

Proposition 11 (Denominator)**.** *Let $F_\alpha = \mathbb{Q}(a)$ and $F_\beta = \mathbb{Q}(b)$. Then*

$$w_2(F_\alpha) = 60 = 2^2 \cdot 3 \cdot 5, \qquad w_2(F_\beta) = 204 = 2^2 \cdot 3 \cdot 17,$$

with $\mathrm{disc}\,F_\alpha = -5^6 \cdot 59 \cdot 3881^2$ *and* $\mathrm{disc}\,F_\beta = 17^{14} \cdot 1597$*. By the identification of the extended Bloch group with K_3^{ind} and the Merkurjev–Suslin torsion formula $|K_3^{\mathrm{ind}}(F)_{\mathrm{tors}}| = w_2(F)$, the order of each class divides the corresponding w_2, so the denominator of* $\mathrm{Re}[\Delta\mathcal{R}]/\pi^2$ *divides*

$$Q = \mathrm{lcm}(60, 204) = 1020.$$

Theorem 12 (Main)**.** $\displaystyle\int_\alpha^\beta \Big(\log x\,\frac{dy}{y} - \log y\,\frac{dx}{x}\Big) = \frac{4\pi^2}{85}.$

Proof. By Theorem **??** and Proposition **??**, $\mathrm{Re}[\Delta\mathcal{R}]/\pi^2$ is a rational number with denominator dividing $Q = 1020$. Two distinct rationals with denominator at most Q differ by at least $1/Q^2$, so such a rational is determined uniquely by any approximation of error less than $1/(2Q^2) = 4.81 \times 10^{-7}$.

Evaluating (**??**) at 1000 bits of working precision from the exact endpoint data gives

$$\frac{\mathrm{Re}[\Delta\mathcal{R}]}{\pi^2} = -0.0470588235294117647058823529411764705882352941176470588235294\ldots$$

with $\big|\mathrm{Re}[\Delta\mathcal{R}]/\pi^2 + \frac{4}{85}\big| = 1.1 \times 10^{-301}$, far below the separation threshold. Rational reconstruction with denominator bound Q therefore returns *uniquely*

$$\frac{\mathrm{Re}[\Delta\mathcal{R}]}{\pi^2} = -\frac{4}{85}.$$

(Consistency: $85 \mid 1020$, with quotient 12.) By (**??**), $I = -\Delta\mathcal{R} = 4\pi^2/85$. □

Corollary 13. $\mathrm{GV}(\rho_\alpha) = \dfrac{74\pi^2}{15}$*, by* (**??**).

6 Reproducing the computations

All computations are in `scripts/`; each is standalone (Sage, or Python with `mpmath`/`SnapPy`).

`falpha2.sage`	derives (??) from $A(a^2, a)$
`confirm.sage`	Lemma **??**: unit-circle counts via g_α, g_β
`exact_proof.sage`	Lemma **??**: the identity $u(1/a) = u(a)$
`mechanism.py`	Lemmas **??**–**??** at every embedding
`filling_term.py`	Lemma **??**: λ^2_{core} real
`orient.py`	Lemma **??**: SnapPy triangulation, four positive tetrahedra
`alpha_component.py`	no embedding realises $\mathrm{Vol}(S^3_{-1/2}(7_2))$
`disc_test.py`	non-discreteness (cyclotomic trace test)
`w2bound.sage`	Proposition **??**: $w_2 = 60,204$
`shapes300.sage`, `pin.sage`	the 301-digit evaluation and reconstruction

References

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